

SUMMER INTERNSHIP COURSE

2024-25

1. **Purpose/Objective:** The NCP51820 is a high-speed half-bridge gate driver designed specifically for driving enhancement mode (e-mode) high electron mobility transistors (HEMTs) and gate injection transistors (GITs) made of gallium nitride (GaN) in off-line, half-bridge power topologies.
2. **Background/Context:** The NCP51820 is a high-speed half-bridge gate driver designed for driving GaN power switches in off-line, half-bridge power topologies, offering advanced features for high-speed switching applications.
3. **Methodology/Approach:** The NCP51820 high-speed half-bridge gate driver uses advanced level shift technology, dedicated voltage regulators, and comprehensive protection features to drive GaN power switches in off-line, half-bridge power topologies.
4. **Findings/Results:** The NCP51820 high-speed half-bridge gate driver offers key features such as high-speed operation, wide common-mode voltage range, stable dV/dt operation, comprehensive protection functions, programmable dead-time control, and high drive capability to enable efficient and reliable driving of GaN power switches in off-line, half-bridge power topologies.
5. **Analysis/Interpretation:** The NCP51820 high-speed half-bridge gate driver provides a robust and efficient solution for driving GaN power switches in off-line, half-bridge power topologies, offering advanced features such as high-speed operation, wide common-mode voltage range, stable dV/dt operation, comprehensive protection functions, programmable dead-time control, and high drive capability.
6. **Conclusion/Recommendation:** The NCP51820 high-speed half-bridge gate driver provides a robust and efficient solution for driving GaN power switches in off-line, half-bridge power topologies, offering advanced features such as high-speed operation, wide common-mode voltage range, stable dV/dt operation, comprehensive protection functions, programmable dead-time control, and high drive capability, making it well-suited for a variety of soft-switching applications.

1. Basic Information

Name of Student	Jagadeeswara Chandra bose.K
Roll Number	AP23122100002
Branch	M.Tech ECE Embedded systems and IOT
Name of Internship station/ Company	Indian Institute of Technology Palakkad.
Location	Latitude: 10.784703 Longitude: 76.653145
Date of Joining	03-06-2024
Address of the Company	Malampuzha Rd, near Gramalakshmi Mudralayam, Kanjikode, Kerala 678623, India.
Project Title	Power Electronic Circuits - Numerical Simulations

Project Purpose (**One sentence only - about 10/12 words**, describing the anticipated change. What is the immediate outcome or direct benefit the project will achieve resulting from the activities and outputs). It should not contain project details which can be described elsewhere on the form.

The NCP51820 high-speed half-bridge gate driver is designed to provide a robust and efficient solution for driving gallium nitride (GaN) power switches in off-line, half-bridge power topologies.

<i>What is the situation/status in the company before the project was given to you?</i>	<i>Indicators of success (evidence: how we will know the purpose (above) has been achieved?)</i>	<i>What is the progress till date?</i>
I am with Fundamental knowledge only when I am starting	Started learning from scratch in a standard format along with PhD scholars and Guide	Learned different types of Standard converters
Zero-knowledge on Calculation	Started learning from standard books	Created Formula based Spreadsheets to optimize the Calculation time and validation in LT spice tool
No experience with Industrial Datasheets	Step by Step process to read the datasheet	Learned to study Data sheet and Manufacture standards

Outputs: Please list here all of the outputs (specific deliverables) you expect the project activities to deliver.

Outputs <i>(The results of project activities. These should be sufficient to achieve the project purpose.)</i>	1. Power Electronics Converters 2. EDA Tools – LT spice and Altium PCB design 3. Datasheets and Spread sheet designing
Main Activities <i>(List the tasks to be done to deliver the outputs.)</i>	1.1 Power Electronics Converters 1.2 LTspice Simulation Software 1.3 Altium Designer for PCB Design 2.1 Power Electronics Circuit Design and Simulations 2.2 NCP51820 Gate Driver (ON Semiconductor) 2.3 GaN3R2-100CBE MOSFET (Nexperia)

Brief Background of the Project

(500 words max. Please include the rationale, the context and relevant/expected work to be conducted in this area)

The NCP51820 is a high-speed half-bridge gate driver designed specifically for driving gallium nitride (GaN) power switches in off-line, half-bridge power topologies, offering features such as fast rise and fall times, wide common mode voltage range, stable dV/dt operation, dedicated gate protection, comprehensive protection functions, programmable dead-time control, and high drive capability to enable efficient and reliable utilization of GaN power switches in modern power electronics designs, particularly in soft-switching applications like full or half-bridge converters, LLC resonant converters, active clamp flyback or forward converters, totem pole PFC circuits, and synchronous rectifier topologies, as well as industrial inverters, motor drives, and AC-to-DC converters.


Signature of Student:

(I confirm that all relevant project-related information has been shared and I agree that I shall work towards the goals set in this form)